

IMMANUVEL ALEX

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EDUCATION

Rajagiri College of Management and Applied Sciences

Bachelor of Computer Application

Kochi, India

Cambrian College

Diploma, Computer System Technician

Sudbury, Canada

INDEPENDENT R&D

Cybersecurity & AI Research

Ernakulam, India | 2025 – Present

Full-time independent research following international work experience in Canada

- Designed, built, and shipped **AegisX v3.3.0** — a 44-module, production-grade AI-powered cybersecurity platform — and **Aegis-Trade**, an autonomous AI trading system, entirely as independent R&D projects.
- Completed **6 Anthropic certifications** (MCP, Claude Code, Subagents, Agent Skills), **Elements of AI (2 ECTS)**, and multiple Google Cloud GenAI credentials during this period.

PROJECTS

AegisX v3.3.0 PHANTOM-X – Python, Rust, React, Flask, Docker, ZeroMQ, Claude API

Mar 2026

github.com/A3lxq/AEGISX-v3.3.0

- Architected a **44-module, 49-page** open-source unified security operations platform from scratch — covering red team offense (24 modules), blue team defense (13 modules), and purple team simulation; **30,339 lines of Python, 0 syntax errors**.
- Designed **AEGIS SHIELD** — a 9-engine real-time threat detection daemon detecting DNS exfiltration, ICMP tunneling, C2 beacon patterns, NTLM relay, and TLS anomalies, each automatically mapped to MITRE ATT&CK techniques.
- Integrated **Claude API** as the AI reasoning core — powering an autonomous pentesting agent, AI-assisted vulnerability discovery from source code, AI spear-phishing generation from OSINT data, and threat intelligence correlation.
- Built production-grade infrastructure: Flask REST API with Bearer auth, Docker stack (Kali + Redis + Nginx), ZeroMQ real-time streaming, SQLite session persistence, and a **781-tool BlackArch library** with AI-guided tool selection.
- Platform replaces **28 commercial tools** (Cobalt Strike, Burp Suite Pro, Qualys VMDR, Recorded Future) — estimated saving of **\$377,000+/year** in enterprise licensing costs.

Aegis-Trade – React Native, Expo, Electron, Claude API, MetaTrader 5, Python

2025 – Present

github.com/A3lxq/Aegis-Trade

- Building an autonomous AI trading platform integrating **TradingView webhooks** and **MetaTrader 5** execution, with Claude API as the core reasoning engine for real-time signal interpretation and trade decision-making.
- Dual-codebase architecture: Android app (React Native/Expo) for mobile monitoring and Windows desktop app (Electron/React) for full control, backtesting, and position management.

Portfolio Website – HTML5, Tailwind CSS, JavaScript, Netlify

2024

- Built a fully responsive portfolio with dark mode, AI chatbot simulation, smooth animations, and cloud deployment via Netlify.

SKILLS

Languages & Frameworks: Python, Rust, JavaScript (ES6), HTML5, CSS3, Tailwind CSS, React, React Native, Flask, FastAPI

Tools & Infrastructure: Docker, ZeroMQ, SQLite, Redis, Git, Linux, Windows, Active Directory, TCP/IP, LAN/WAN, DNS, DHCP

Cybersecurity: Ethical Hacking, VAPT, MITRE ATT&CK, Threat Detection, Penetration Testing, OSINT, Malware Analysis, Incident Response

AI & Automation: Claude API, MCP (Intro + Advanced), AI Subagents, Agent Skills, Prompt Engineering, LLM Integration, Workflow Automation

CERTIFICATIONS

Anthropic: Introduction to MCP, MCP Advanced Topics, Claude Code 101, Claude Code in Action, Introduction to Subagents, Introduction to Agent Skills

University of Helsinki / MinnaLearn: Elements of AI (2 ECTS)

Google Cloud Skills: Introduction to Generative AI, Gen AI: Beyond the Chatbot, Gen AI Agents: Transform Your Organization

HP LIFE: Data Science & Analytics, Introduction to Cybersecurity Awareness

LANGUAGES

English (Fluent) | Malayalam (Native) | Hindi | Punjabi